

Imperial College London

BSc/MSci EXAMINATION Summer 2022

This paper is also taken for the relevant Examination for the Associateship

MRES - STATISTICAL METHODS FOR EXPERIMENTAL PHYSICS

For MLBD MRes Students

July 6th, 2022: 10:00 to 12:00

Answer all FOUR questions

Marks shown on this paper are indicative of those the Examiners anticipate assigning.

General Instructions

Complete the front cover of each of the 4 answer books provided.

If an electronic calculator is used, write its serial number at the top of the front cover of each answer book.

USE ONE ANSWER BOOK FOR EACH QUESTION.

Enter the number of each question attempted in the box on the front cover of its corresponding answer book.

Hand in 4 answer books even if they have not all been used.

You are reminded that Examiners attach great importance to legibility, accuracy and clarity of expression.

1. (i) The binomial distribution describes the probability to observe k successful outcomes in n independent trials,

$$P(k; n, p) = \frac{n!}{k!(n-k)!} p^k (1-p)^{n-k}$$

where p is the probability that a single outcome is successful.

- (a) Show that the expectation value of the binomial distribution is np

ANSWER: We can express the number of successful trials k as the sum over the outcomes X_i , $k = \sum_{i=1}^n X_i$. The expectation of k is given by,

$$E[k] = E\left[\sum_{i=1}^n X_i\right] = \sum_{i=1}^n E[X_i] = np$$

- (b) Show that the variance of a binomial distribution is $np(1-p)$. You can use the fact that the variance of the sum of independent variables is the same as the sum of the variances of the independent variables.

ANSWER: Similar to above,

$$V[k] = V\left[\sum_{i=1}^n X_i\right] = \sum_{i=1}^n V[X_i] = np(1-p)$$

[10 marks]

- (ii) The Poisson distribution describes the probability to observe k successful outcomes (events) when those outcomes are rare. Formally, the Poisson distribution arises as the result of a binomial distribution for which $p \rightarrow 0$ as $n \rightarrow +\infty$ such that $\lambda = np$ is constant. The Poisson probability distribution is given by,

$$P(k; \lambda) = \frac{\lambda^k}{k!} e^{-\lambda}$$

- (a) Using the method of moments, find an estimator for the λ parameter $\hat{\lambda}$ assuming an observed number of events k .

ANSWER: To obtain the estimator for the parameter λ , we can equate the sample mean with the expectation value of the Poisson.

$$E[k] = \bar{k}$$

For a single observation of k events, the sample mean will just be $\bar{k} = k$. Similarly, the expectation value of the poisson distribution is λ and hence by setting the expectation value equal to the sample mean, we say that the estimator $\hat{\lambda} = k$

- (b) Show that the maximum likelihood estimator for λ is the same as the estimator found in the previous question.

ANSWER: From the definition of the Poisson, we find

$$L(\lambda) = \frac{\lambda^k}{k!} e^{-\lambda}$$

[This question continues on the next page ...]

and so taking a negative log, we get,

$$q = -\ln L(\lambda) = \lambda - k \ln \lambda + \ln(k!)$$

The maximum likelihood estimator for λ is found by minimizing q (equivalent to maximizing L) with respect to λ .

$$\frac{dq}{d\lambda}\bigg|_{\lambda=\hat{\lambda}} = 0 \implies 1 - \frac{k}{\hat{\lambda}} = 0$$

and so,

$$\hat{\lambda} = k$$

- (c) Calculate the third central moment of the Poisson distribution. Remember that the n -th central moment of the distribution of a discrete random variable k is defined as,

$$\nu_n = E[(k - E[k])^n],$$

where $E[\cdot]$ is the expectation value. You may use the following identities;

$$(x^2 + x)e^x = \sum_{k=0}^{+\infty} k^2 \frac{x^k}{k!}$$

$$xe^x = \sum_{k=0}^{+\infty} k \frac{x^k}{k!}$$

$$e^x = \sum_{k=0}^{+\infty} \frac{x^k}{k!}$$

ANSWER: The definition of the *third central moment* of a discrete random variable k with probability distribution $P(k)$ was defined in lectures to be,

$$E[(k - E[k])^3] = \sum_k (k - E[k])^3 P(k).$$

For the case where $p(k)$ is a Poisson distribution with parameter λ this becomes,

$$E[(k - E[k])^3] = \sum_{k=0}^{\infty} (k - \lambda)^3 \lambda^k \frac{e^{-\lambda}}{k!},$$

where we used the fact that $E[k] = \lambda$ for the Poisson case, as shown in the lectures and used in the previous questions.

Expanding the brackets.

$$\begin{aligned}
 E[(k - E[k])^3] &= \sum_{k=0}^{\infty} (k - \lambda)^3 \lambda^k \frac{e^{-\lambda}}{k!} \\
 &= \sum_{k=0}^{\infty} k^3 \lambda^k \frac{e^{-\lambda}}{k!} \\
 &+ 3\lambda^2 \sum_{k=0}^{\infty} k \lambda^k \frac{e^{-\lambda}}{k!} \\
 &- 3\lambda \sum_{k=0}^{\infty} k^2 \lambda^k \frac{e^{-\lambda}}{k!} \\
 &- \lambda^3 \sum_{k=0}^{\infty} \lambda^k \frac{e^{-\lambda}}{k!}.
 \end{aligned}$$

The last term is just $\lambda^2 \sum_{k=0}^{\infty} p(k) = \lambda^2$ as the total probability ($\sum_{k=0}^{\infty} p(k)$) must equal 1. For the 2nd and 3rd terms, we can use the identities provided to simplify them so that we are left with,

$$\begin{aligned}
 E[(k - E[k])^3] &= \sum_{k=0}^{\infty} (k - \lambda)^3 \lambda^k \frac{e^{-\lambda}}{k!} \\
 &= \sum_{k=0}^{\infty} k^3 \lambda^k \frac{e^{-\lambda}}{k!} \\
 &+ 3\lambda^2 \times \lambda \\
 &- 3\lambda \times (\lambda^2 + \lambda) \\
 &- \lambda^3.
 \end{aligned}$$

We notice that,

$$\begin{aligned}
 \sum_{k=0}^{\infty} k^3 \lambda^k \frac{e^{-\lambda}}{k!} &= 0 + \sum_{k=1}^{\infty} k^3 \lambda^k \frac{e^{-\lambda}}{k!} \\
 &= \sum_{k=1}^{\infty} (k \times k^2) (\lambda \times \lambda^{k-1}) \frac{e^{-\lambda}}{k \times (k-1)!} \\
 &= \lambda \sum_{k=1}^{\infty} k^2 \lambda^{k-1} \frac{e^{-\lambda}}{(k-1)!} \\
 &= \lambda \sum_{p=0}^{\infty} (p+1)^2 \lambda^p \frac{e^{-\lambda}}{p!},
 \end{aligned}$$

where we've made the substitution $k \rightarrow p + 1$ in the last line. Now we can expand the brackets to get,

$$\lambda \sum_{p=0}^{\infty} (p+1)^2 \lambda^p \frac{e^{-\lambda}}{p!} = \lambda \left(\sum_{p=0}^{\infty} p^2 \lambda^p \frac{e^{-\lambda}}{p!} + \sum_{p=0}^{\infty} 2p \lambda^p \frac{e^{-\lambda}}{p!} + \sum_{p=0}^{\infty} \lambda^p \frac{e^{-\lambda}}{p!} \right),$$

and once again we can simplify using the provided identities,

$$\lambda \sum_{p=0}^{\infty} (p+1)^2 \lambda^p \frac{e^{-\lambda}}{p!} = \lambda (\lambda^2 + \lambda + 2\lambda + 1) = \lambda^3 + 3\lambda^2 + \lambda,$$

and finally we can substitute this back into the expression for the third moment to get to,

$$\begin{aligned} E[(k - E[k])^3] &= \sum_{k=0}^{\infty} (k - \lambda)^3 \lambda^k \frac{e^{-\lambda}}{k!} \\ &= \lambda^3 + 3\lambda^2 + \lambda + 3\lambda^3 - 3\lambda^3 - 3\lambda^2 - \lambda^3 \\ &= \lambda, \end{aligned}$$

so the third central moment of a Poisson distribution is λ .

[15 marks]

[Total 25 marks]

2. The Gaussian probability distribution for a continuous random variable x is given by,

$$f(x; \mu, \sigma) = \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{x - \mu}{\sigma} \right)^2}.$$

(i) Imagine an experiment that measures observations of x . Suppose this experiment measures a sequence of n independent observations $\{x_1, x_2, \dots, x_n\}$, assumed to be distributed as Gaussian random variables with the same μ and σ parameters.

(a) Write down the likelihood $L(\mu, \sigma)$ for the sequence of observations

ANSWER: The likelihood can just be written using the probability density provided as,

$$L(\mu, \sigma) = \prod_{i=1}^n f(x_i; \mu, \sigma)$$

(b) Define the quantity $q = -\ln L(\mu, \sigma)$. Use this to determine the maximum likelihood estimators $\hat{\mu}$ and $\hat{\sigma}$.

ANSWER: Taking the negative log of the likelihood, we have

$$q = -\sum_{i=1}^n \left[-\ln(2\pi\sigma) - \frac{1}{2} \left(\frac{x_i - \mu}{\sigma} \right)^2 \right].$$

Again, to find the maximum likelihood estimates, we find the minimum of q . This occurs when,

$$\frac{\partial q}{\partial \mu} \Big|_{\mu=\hat{\mu}} = 0$$

and

$$\frac{\partial q}{\partial \sigma} \Big|_{\sigma=\hat{\sigma}} = 0$$

From the first equation, we get,

$$0 = \sum_{i=1}^n \frac{1}{\sigma} \left(\frac{x_i - \hat{\mu}}{\sigma} \right) = \sum_{i=1}^n x_i - n\hat{\mu} \implies \hat{\mu} = \bar{x}$$

since $\sigma > 0$ and $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$.

From the second equation, we get,

$$0 = \frac{n}{\hat{\sigma}} - \frac{1}{\hat{\sigma}^3} \sum_{i=1}^n (x_i - \mu)^2 = n - \frac{1}{\hat{\sigma}^2} \sum_{i=1}^n (x_i - \mu)^2 \implies \hat{\sigma} = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2}.$$

Since the two equations must hold at the minimum point, we substitute $\mu = \hat{\mu}$ in the second one, so that,

$$\hat{\sigma} = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2}.$$

[This question continues on the next page ...]

[10 marks]

- (ii) A charged particle tracker is designed to measure the direction of charged particles using 10 layers of silicon detectors. The tracker is designed such that at each layer position (x_i) the point at which the particle traverses the silicon along the layer (y_i) is measured with some accuracy. Take a look at Figure 1. Each of the data points represents the measured value of the particle position y_i along the layer x_i . The measured values y_i are assumed to be Gaussian random variables with variance σ^2 , while the silicon layer positions x_i are known exactly. The line in the figure represents a hypothesis for the charged particle track, which can be represented by the equation of a straight line,

$$y = mx + c$$

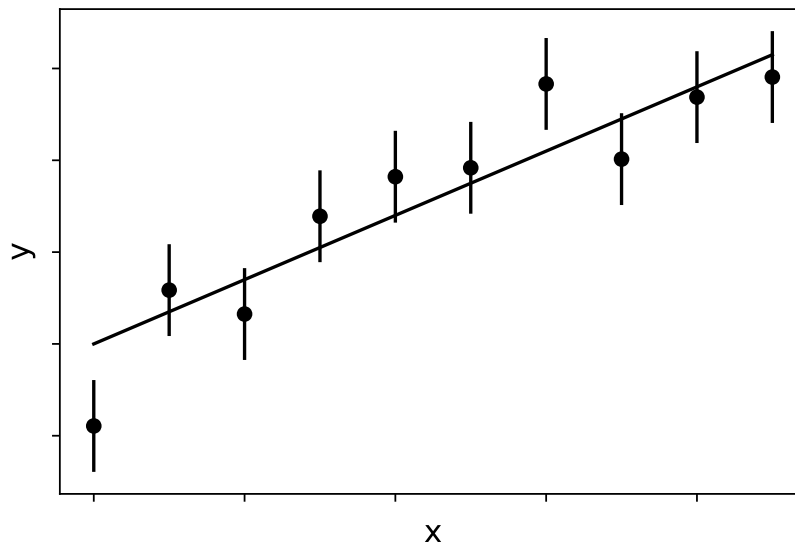


Figure 1

The joint probability density for n measurements can be expressed as a multivariate Gaussian distribution,

$$f(\underline{y}; \underline{\mu}, \sigma) = f(\underline{y}; m\underline{x} + c, \sigma) = \frac{1}{\sqrt{(2\pi)^n \sigma^{2n}}} e^{-\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - (mx_i + c))^2}$$

where in this case $\underline{y} = \{y_1, y_2, \dots, y_{10}\}$ and $n = 10$.

- (a) Write down the likelihood $L(m, c)$ and $q = -\ln L$ for the measured positions as functions of the track parameters m and c

ANSWER: The likelihood can just be written using the probability density

[This question continues on the next page ...]

provided and treating the n measurements as a single observation vector $\underline{x}$. In this case, the likelihood is just,

$$L(m, c) = \frac{1}{\sqrt{(2\pi)^n \sigma^{2n}}} e^{-\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - (mx_i + c))^2}$$

and

$$q = \frac{n}{2} \ln(2\pi) + n \ln \sigma + \frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - mx_i - c)^2$$

- (b) Find the maximum likelihood estimators $\hat{m}$ and $\hat{c}$

ANSWER: The maximum likelihood estimators can be obtained from minimising q from the previous question.

$$\frac{\partial q}{\partial m} = 0 \implies m \sum_{i=1}^n x_i^2 + c \sum_{i=1}^n x_i - \sum_{i=1}^n y_i x_i = 0$$

and

$$\frac{\partial q}{\partial c} = 0 \implies \sum_{i=1}^n y_i - m \sum_{i=1}^n x_i - nc = 0$$

Notice that $\sum_{i=1}^n y_i = \bar{y}$ and $\sum_{i=1}^n x_i = \bar{x}$. The second equation now reads,

$$n\bar{y} - mn\bar{x} - nc = 0 \rightarrow c = \bar{y} - m\bar{x}.$$

Now we substitute the expression for c back into the first equation and we get,

$$m \sum_{i=1}^n x_i^2 + n\bar{x}(\bar{y} - m\bar{x}) - \sum_{i=1}^n x_i y_i = 0$$

which we can rewrite as

$$m \left(\sum_{i=1}^n x_i^2 - n\bar{x}^2 \right) + \left(n\bar{x}\bar{y} - \sum_{i=1}^n x_i y_i \right) = 0$$

Both terms in the brackets can be expressed as sums of products of differences,

$$\sum_{i=1}^n x_i^2 - n\bar{x}^2 = \sum_{i=1}^n (x_i - \bar{x})(x_i + \bar{x}), \quad \sum_{i=1}^n x_i y_i - n\bar{x}\bar{y} = - \sum_{i=1}^n (y_i - \bar{y})(x_i + \bar{x})$$

so we can write finally that

$$\hat{m} = \frac{\sum_{i=1}^n (y_i - \bar{y})(x_i + \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})(x_i + \bar{x})}$$

and

$$\hat{c} = \bar{y} - \bar{x} \left(\frac{\sum_{i=1}^n (y_i - \bar{y})(x_i + \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})(x_i + \bar{x})} \right)$$

These can be expressed in terms of sample means of x and y as above or left with the summation terms.

[This question continues on the next page ...]

- (c) Describe how you would determine a 68% confidence region for the parameter space m vs c from the measurements using the Neyman construction and a profile likelihood ratio test-statistic.

ANSWER: The answer should contain the following set of steps:

1. Choose a test statistic t for the data. For example, this could be the based on log likelihood ratio $t = -2 \ln \left(\frac{L(\tau)}{L(\hat{\tau})} \right)$
 2. Calculate the value of the test statistic for the data t_{obs} , which should depend on m and c .
 3. For a grid of m and c values derive the distribution $f(t; m, c)$ of the test-statistic assuming m and c are the true values. Choose a procedure to construct 68% of the distribution in t for each value of m, c . Any well motivated procedure is ok, for example the likelihood ratio ordering principle is ok, or if the test-statistic is the one based on the log likelihood ratio, largest values could be included first.
 4. For each value of m and c determine whether the observed value t_{obs} lies inside the 68% region or not. This could be based on calculating a p-value $p = \int_{t_{obs}}^{\infty} f(t; m, c) dt$ and choosing grid points (m, c) for which $p > 1 - 0.68 = 0.32$.
 5. The union of those chosen points forms the 68% confidence region in the m, c plane.
- (d) The quality of the track fit is often thought of in terms of the ‘goodness-of-fit’. The appropriate test statistic is defined as,

$$t = -2 \ln \left(\frac{f(\underline{y}; \hat{m}, \hat{c}, \sigma)}{f(\underline{y}; \underline{y}, \sigma)} \right),$$

where $f(\underline{y}; \underline{y}, \sigma)$ is the ‘saturated model’ for which the mean vector - $\underline{\mu}$ - of the multivariate Gaussian is equal to the observed values $\underline{y}$. This test statistic is distributed as a chi-square function with 10 degrees of freedom. Write down t and describe how you would perform a hypothesis test to determine whether the fitted track (the straight line for $m = \hat{m}$ and $c = \hat{c}$) is a good description of the data.

ANSWER: t can be found by plugging the density f into the definition given for t . The first thing to notice is that

$$f(\underline{y}; \underline{y}, \sigma) = \frac{1}{\sqrt{(2\pi)^n \sigma^{2n}}}$$

since the term in the exponent yields 1. Then plugging into t , we have

$$t = -2 \ln \left[\frac{\frac{1}{\sqrt{(2\pi)^n \sigma^{2n}}} e^{-\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - (\hat{m}x_i + \hat{c}))^2}}{\frac{1}{\sqrt{(2\pi)^n \sigma^{2n}}}} \right] = \frac{1}{\sigma^2} \sum_{i=1}^n (y_i - (\hat{m}x_i + \hat{c}))^2$$

Note, it is not correct to substitute expressions for $\hat{m}$ and $\hat{c}$ from the previous question since this is the general test statistic not specifically the observed value. The answer for the second part should explain the basics of a hypothesis test, namely,

1. Given the test statistic, choose a threshold for the size of the test α , eg $\alpha = 0.05$.
2. From the form of the test statistic, the student should know that larger values of the statistic indicate worse agreement between the data and the null hypothesis H_0 defined by setting $m = \hat{m}$ and $c = \hat{c}$ from the observed data. The distribution of the test statistic can be generated with random toys but since the question says how it will be distributed, we can just use that to determine a t_α for which a fraction α of the distribution of t is larger than t_α or $P(t > t_\alpha | H_0) = 0.05$.
3. Evaluate the observed value of the test statistic t_{obs} . If $t_{obs} > t_\alpha$ then we reject the null hypothesis - the fit is not of good quality.

[15 marks]

[Total 25 marks]

3. (i) (a) In the context of Bayesian parameter inference, and with reference to Bayes' theorem, identify the terms *likelihood*, *prior*, *posterior*, *evidence*.

ANSWER: Bayes theorem:

$$p(\theta|d) = \frac{p(d|\theta)p(\theta)}{p(d)}.$$

Likelihood is $p(d|\theta)$, prior is $p(\theta)$, posterior is $p(\theta|d)$, evidence is $p(d)$. [2]

- (b) What is the difference between the likelihood and the sampling distribution?

ANSWER: They both have the same functional form $p(d|\theta)$, but the sampling distribution is a function of d , with θ fixed, whereas the likelihood treats it as a function of θ at fixed d . [3]

[5 marks]

- (ii) A machine is designed to measure the length μ of a crucial component in a medical device. The measurement error of the machine is characterised by a gaussian distribution with zero mean and unknown standard deviation σ . An experiment is performed to infer both μ and σ , by making a set $\{x\}$ of repeated independent measurements of the length: $x_i; i = 1, \dots, n$.

- (a) What is the posterior distribution of the model parameters μ and σ (up to a normalisation constant which you do not need to calculate)? You should assume that the prior factorises into $\pi(\mu)\pi(\sigma)$.

ANSWER:

$$\begin{aligned} p(\mu, \sigma|\{x\}) &\propto p(\{x\}|\mu, \sigma)\pi(\mu, \sigma) \\ &\propto \pi(\mu)\pi(\sigma) \times \prod_{i=1}^n \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{(x_i - \mu)^2}{2\sigma^2}\right] \end{aligned} \quad (1)$$

[3]

- (b) Choose a suitable prior $\pi(\mu)$ and justify your answer.

ANSWER: μ is a location parameter so a uniform prior is most suitable. [2]

- (c) Two possibilities for a prior $\pi(\sigma)$ suggest themselves: $\pi(\sigma) \propto \sigma^{-q}$, with $q = 0$ and $q = 1$. $q = 0$ is evidently a uniform prior, but provide a justification for the choice $q = 1$.

ANSWER: σ is a scale parameter, and is strictly positive, so a Jeffreys prior $\pi(\sigma) \propto 1/\sigma$ is justified (or, equivalently, the prior on $\ln(\sigma)$ is uniform). [2]

[7 marks]

- (iii) For now, keep q general.

This experiment is designed to determine the variance of the measurements made by the machine. Show that the marginal posterior distribution of σ is

$$p(\sigma|\{x\}) \propto \frac{1}{\sqrt{n}\sigma^{n+q-1}} \exp\left(-\frac{n\sigma_x^2}{2\sigma^2}\right)$$

[This question continues on the next

where

$$\sigma_x^2 \equiv \frac{1}{n} \sum_{i=1}^n x_i^2 - \bar{x}^2$$

and $\bar{x}$ is the average of the x_i .

ANSWER:

$$\begin{aligned} p(\sigma|\{x\}) &= \int_{-\infty}^{\infty} d\mu p(\mu, \sigma|\{x\}) \\ &\propto \int_{-\infty}^{\infty} d\mu \pi(\mu) \pi(\sigma) \times \prod_{i=1}^n \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{(x_i - \mu)^2}{2\sigma^2}\right] \\ &\propto \frac{1}{\sigma^{q+n}} \int_{-\infty}^{\infty} d\mu \exp\left[-\sum_{i=1}^n \frac{(x_i - \mu)^2}{2\sigma^2}\right] \\ &\propto \frac{1}{\sigma^{q+n}} \int_{-\infty}^{\infty} d\mu \exp\left[-\sum_{i=1}^n \frac{x_i^2}{2\sigma^2} + \frac{\mu}{\sigma^2} \sum_{i=1}^n x_i - \frac{n\mu^2}{2\sigma^2}\right] \\ &\propto \frac{1}{\sigma^{q+n}} \int_{-\infty}^{\infty} d\mu \exp\left[-\sum_{i=1}^n \frac{x_i^2}{2\sigma^2} - \frac{n(\mu - \bar{x})^2}{2\sigma^2} + \frac{n\bar{x}^2}{2\sigma^2}\right] \end{aligned} \quad (2)$$

Integrating w.r.t. μ gives $\sqrt{2\pi\sigma^2/n}$, hence

$$p(\sigma|\{x\}) \propto \frac{1}{\sqrt{n}\sigma^{q+n-1}} \exp\left(-\frac{n\sigma_x^2}{2\sigma^2}\right).$$

[8 marks]

- (iv) On the basis of the marginal distribution of σ , find the most probable value of σ , in terms of σ_x , for the cases $q = 0$ and $q = 1$.

ANSWER: Easiest to maximise the log of the posterior:

$$\begin{aligned} \frac{d \ln p(\sigma|\{x\})}{d\sigma^2} &= 2\left(\frac{-n-q+1}{2\sigma}\right) + 2\left(\frac{n\sigma_x^2}{2\sigma^3}\right) = 0 \\ \Rightarrow \sigma^2 &= \frac{n}{n+q-1} \sigma_x^2. \end{aligned} \quad (3)$$

i.e. $\sigma = \sigma_x$ ($q = 1$) or $\sqrt{\frac{n}{n-1}} \sigma_x$ ($q = 0$).

[5 marks]

[Total 25 marks]

4. Two models of the Universe are proposed. In one (Model 0), the Universe is flat, and the curvature of the Universe k is zero. In the other (Model 1), the curvature k may be non-zero. A measurement d of the curvature is made, which is subject to a gaussian error of zero mean and known variance σ^2 .

- (i) (a) Write down the likelihood $p(d|k)$ for Model 0 and for Model 1.

ANSWER:

$$\begin{aligned} p(d|k=0) &= \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{d^2}{2\sigma^2}\right] && \text{Model 0} \\ p(d|k) &= \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{(d-k)^2}{2\sigma^2}\right] && \text{Model 1} \end{aligned} \quad (1)$$

[3]

- (b) In Model 1, assuming a gaussian prior $\pi(k)$ with zero mean and variance Σ^2 , what is the (unnormalised) posterior, $p(k|d)$?

ANSWER:

$$p(k|d) \propto p(d|k)\pi(k) \propto \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{(d-k)^2}{2\sigma^2}\right] \times \frac{1}{\sqrt{2\pi}\Sigma} \exp\left[-\frac{k^2}{2\Sigma^2}\right]$$

[3]

[6 marks]

- (ii) (a) Show that the most probable (maximum *a posteriori*, or MAP) value of k in Model 1 is

$$k_{\text{MAP}} = \frac{d\Sigma^2}{\sigma^2 + \Sigma^2}.$$

ANSWER: Maximising $\ln p(k|d) = \text{cst.} - \frac{(d-k)^2}{2\sigma^2} - \frac{k^2}{2\Sigma^2}$ for convenience, we find

$$0 = \frac{d-k}{\sigma^2} - \frac{k}{\Sigma^2} \Rightarrow k_{\text{MAP}} \left(\frac{1}{\sigma^2} + \frac{1}{\Sigma^2} \right) = \frac{d}{\sigma^2}$$

and result follows. [3]

- (b) What is the variance of the posterior in Model 1?

ANSWER: Rearranging the exponent, we find

$$p(k|d) \propto \exp\left[-\frac{(k-k_{\text{MAP}})^2}{2(\sigma^2 + \Sigma^2)}\right] \exp\left[-\frac{d^2}{2\sigma^2}\right] \exp\left[\frac{k_{\text{MAP}}^2}{2(\sigma^2 + \Sigma^2)}\right]$$

The last two terms do not depend on k , so the posterior is a gaussian in k with variance $\sigma^2 + \Sigma^2$, by inspection (or you can do the integral, but it will be time-consuming). [3]

[6 marks]

[This question continues on the next page ...]

- (iii) Find an expression for the Bayesian Evidence $p(d|M_1)$ for Model 1. You may use the result $\int_{-\infty}^{\infty} \exp(bz - cz^2/2) dz = \sqrt{2\pi/c} \exp[b^2/(2c)]$.

ANSWER: Reinserting the prefactors in the likelihood and the prior, the evidence is the integral

$$\begin{aligned} Z_1(d) &= \frac{1}{2\pi\sigma\Sigma} \int_{-\infty}^{\infty} \exp\left[-\frac{(d-k)^2}{2\sigma^2}\right] \exp\left[-\frac{k^2}{2\Sigma^2}\right] dk \\ &= \frac{1}{2\pi\sigma\Sigma} \int_{-\infty}^{\infty} \exp\left[-\frac{k^2}{2}\left(\frac{1}{\sigma^2} + \frac{1}{\Sigma^2}\right) + \frac{dk}{\sigma^2} - \frac{d^2}{2\sigma^2}\right] dk \end{aligned} \quad (2)$$

Using the given result, with $b = d/\sigma^2$ and $c = 1/\sigma^2 + 1/\Sigma^2$, we get

$$Z_1(d) = \frac{1}{\sqrt{2\pi(\sigma^2 + \Sigma^2)}} \exp\left[-\frac{d^2}{2(\sigma^2 + \Sigma^2)}\right].$$

[5 marks]

- (iv) (a) Model 0 has no free parameters. Argue that in this case the Bayesian Evidence $p(d|M_0)$ is just the likelihood.

ANSWER: The evidence is $p(d)$, which is simple if there are no parameters; it is the likelihood $p(d|k=0)$:

$$Z_0(d) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{d^2}{2\sigma^2}\right].$$

[1]

- (b) Find an expression for the Bayes factor

$$B_{01} \equiv \frac{p(d|M_0)}{p(d|M_1)}.$$

ANSWER: Taking the ratio of the evidences:

$$\begin{aligned} B_{01} \equiv \frac{p(d|M_0)}{p(d|M_1)} &= \frac{\sqrt{\sigma^2 + \Sigma^2}}{\sigma} \frac{\exp\left[-\frac{d^2}{2\sigma^2}\right]}{\exp\left[-\frac{d^2}{2(\sigma^2 + \Sigma^2)}\right]} \\ &= \frac{\sqrt{\sigma^2 + \Sigma^2}}{\sigma} \exp\left[-\frac{d^2 \Sigma^2}{2\sigma^2(\sigma^2 + \Sigma^2)}\right] \end{aligned} \quad (3)$$

[3]

- (c) How far away does d have to be from zero for the posterior probability of the two models to be the same? State any further assumptions you make.

ANSWER: Assuming equal prior model probability, the posterior probability ratio is just the Bayes factor. This is unity when $\ln B_{01} = 0$, i.e.

$$0 = \frac{1}{2} \ln\left(\frac{\sigma^2 + \Sigma^2}{\sigma^2}\right) - \frac{d^2 \Sigma^2}{2\sigma^2(\sigma^2 + \Sigma^2)}$$

[This question continues on the next

Hence

$$d^2 = \frac{\sigma^2(\sigma^2 + \Sigma^2)}{\Sigma^2} \ln\left(\frac{\sigma^2 + \Sigma^2}{\sigma^2}\right).$$

[4]

[8 marks]

[Total 25 marks]